ResumeLens

Formal models for structured resume evidence

LeemStarTeam | Discrete Structures III | Integrative Task 1

Can formal models normalize aliases and recognize several qualification patterns?

1 EXTRACT — Regex: raw terms and records → **2 NORMALIZE** — FST: aliases to canonical tokens → **3 RECOGNIZE** — DFA: independent profile patterns → **4 VALIDATE** — textX + rules -> Markdown

Problem and method

Aliases can describe the same technology. Candidates may fit several profiles.

We separate evidence, normalization, recognition and validation using formal models.

Models and profile languages

FST: (Q, Sigma, Gamma, delta, omega, q0, F). Complete aliases are input symbols; canonical terms are output symbols.

DFA: (Q, Sigma, delta, q0, F). One target per state/symbol; no epsilon transitions.

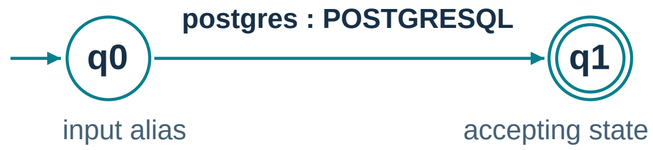
Profile	States	Policy
ML Engineer	**7**	Ordered
Data Architect	**9**	Ordered
Full Stack	**32**	Coverage
Cybersecurity	**16**	Coverage

ML/Data Architect use sorted category tokens. Full Stack/Cybersecurity remember covered categories.

AND across required categories; OR among alternatives. Retain every satisfied profile.

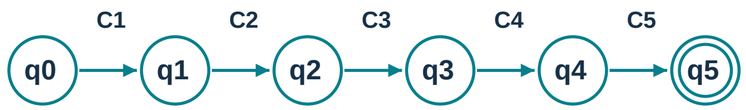
Alias translation: actual FST

Postgres is extracted unchanged. Lexical preparation supplies the atomic symbol **postgres** to the shared skill FST.



The API emits [POSTGRESQL]. Reset q0 for each term. Unknown aliases have no translation.

ML: accepted and rejected traces



C1	PYTHON	q0 -> q1
C2	PANDAS	q1 -> q2
C3	TENSORFLOW	q2 -> q3
C4	POSTGRESQL	q3 -> q4
C5	GIT	q4 -> q5

All five categories: q5 -> ACCEPTED

Without Git: q4 -> REJECTED

Sorting makes resume order independent.

Measured verification results

240 **tests passed**

98 unittest + 142 function cases

Valid candidate exports — 8

Incomplete patterns rejected — 4

Checks compare model tables and cover aliases, missing categories, boundaries, PDF input and UI state.

Conclusion and limits

Conclusion: models give reproducible normalization and multi-profile recognition within the declared input language.

Acceptance is not exportability. Missing records or explicit area levels can block the candidate's export.

Regex does not interpret negation or verify claims. Scanned PDFs require OCR outside this project.

No real-world hiring accuracy was measured. Categories and bounds are team policies, not professional standards.

References and reproducibility

Romero (2021), Pyformlang. DOI: 10.1145/3408877.3432464

Dejanovic et al. (2017), TextX. DOI: 10.1016/j.knosys.2016.10.023

Published models + fictional fixtures + test suite

github.com/emmanue1909/LeemStarTeam